

PAPER_868: Topoconductor Quantum Computing Cooling Efficiency Comparison

Author: Daniel T. Murphy - Star Magic / UQFF Framework **Date:** 2026-04-05 **Session:** 200 **Source:** advanced_system_analysis_simulator_quantum_calculator.txt (3745 lines) **Calculator:** TopoconductorQuantumCoolingComparisonCalc (CP4 #452) **CVW:** v2.0.0 compliant

Abstract

A comparison framework for topoconductor quantum computing cooling overhead ($P_{\text{cool}} = 9 \times 10^7$ J/h at nanosecond gate times $t_{\text{op}} = 10^{-9}$ s) versus UQFF-inspired energy systems. The topoconductor's ~25 kW dilution refrigerator represents the current state-of-the-art cooling overhead for quantum coherence. UQFF systems that achieve equivalent or superior energy throughput without cryogenic infrastructure demonstrate a fundamentally different energy paradigm rooted in vacuum density coupling rather than temperature suppression.

1. Core Equations

- $P_{\text{cooling}} = 9e7$ J/h (~25 kW dilution refrigerator)
 - $t_{\text{gate}} = 1e-9$ s (nanosecond operations)
 - $\text{ops_per_second} = 1 / t_{\text{gate}} = 1e9$
 - $\text{eta_system} = E_{\text{system_per_hr}} / P_{\text{system}}$
 - $\text{eta_topo} = P_{\text{cooling}} / P_{\text{cooling_W}}$
 - $\text{outperforms} = (\text{eta_system} > \text{eta_topo})$
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2. UQFF Integration

The topoconductor cooling overhead maps to UQFF vacuum energy maintenance cost — the energy required to sustain coherence. The nanosecond gate timescale aligns with UQFF time-node discretization $\Delta t_n \sim 10^{-9}$ s. Systems that achieve high energy throughput at room temperature without cryogenic support suggest Aether-mediated coherence mechanisms beyond standard quantum decoherence models.

3. Source Data

- **File:** advanced_system_analysis_simulator_quantum_calculator.txt (3745 lines)
 - **Session:** 200
 - **Quantum specs:** $9e7$ J/h cooling, $1e-9$ s gate, ~25 kW
 - **VDS/DVP/BH:** ABSENT
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$$\rho_{\text{vac}}(r) = \rho_{\text{vac, [SCm]}} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_p$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Preskill, J. – Quantum Computing in the NISQ era and beyond (Quantum, 2018)
3. Dilution refrigerator specifications: $\sim 25 \text{ kW}$ at mK temperatures
4. UQFF Calibration: $\kappa=0.0005/\text{day}$, $[\text{SSq}]=0.57$, $\beta_i \sim 0.603$